

Wake County Sample — Shopping Center / Retail



Traffic Impact Study

Prepared August 8, 2026 · 35.8564, -78.5703

SIS sample draft — full tier

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) evaluates the anticipated traffic impacts of the proposed Wake County Sample — Shopping Center / Retail within Raleigh-Durham CSA, North Carolina (City of Raleigh / Wake County). The analysis follows the structure of the NCDOT Policy on Street and Driveway Access to North Carolina Highways (July 2003) Chapter 5 and the NCDOT Congestion Management Capacity Analysis Guidelines (Standards, March 2022; Best Practices, November 20, 2024). 2 study intersections within a 0.50-mile radius are analyzed under Existing, No-Build, and Build conditions.

2	0	0	6,800
INTERSECTIONS	LOS DROPS	AT LOS E/F	DAILY TRIPS (RAW)

1.0 INTRODUCTION

Project name	Wake County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	35.8564°, -78.5703°
Build-out (horizon) year	2027
Controlling jurisdiction	City of Raleigh / Wake County
MPO	CAMPO (Capital Area Metropolitan Planning Organization)
Programmed projects source	NCDOT 2026-2035 STIP (adopted July 2025), as amended

1.1 TIA Applicability

Raleigh Street Design Manual Section 7.1.3 triggers a TIA at: 150 or more total peak-hour trips; 100 or more peak-hour trips where primary access is on a two-lane roadway; more than 100 peak-direction trips; or 3,000 or more daily trips. Raleigh UDO Section 8.2.2 (Infrastructure Sufficiency — Streets) governs; Sec. 8.2.2.E.5 requires a traffic mitigation plan evaluated by the Transportation Director.

The proposed development's estimated raw trip generation of 6,800 vehicles per day meets or exceeds the 3,000 vpd threshold at which NCDOT may require a TIA (no reductions are applied in the threshold test).

1.2 Scoping

Effective 7/1/2024, Raleigh DOT scopes the TIA jointly with the applicant team and NCDOT where applicable; reviews are performed by the City's on-call consultant with a fee of \$5,500 per submittal (\$1,000 per addendum). Rezoning TIAs compare maximum current-entitlement trips against maximum proposed-entitlement trips. The background growth factor and the list of approved background developments must be agreed in the Approved Scoping Document prior to TIA submittal; a placeholder for that document is carried in the Appendix.

2.0 METHODOLOGY

NCDOT requires HCM-based analysis; the Department currently utilizes Synchro 11 (with mandatory SimTraffic simulation, minimum ten runs), HCS for freeway facilities, and Sidra for roundabouts. This screening analysis applies HCM-consistent computation; Synchro 11-compatible models and SimTraffic outputs are to be provided at formal submittal.

Peak periods	AM and PM weekday peak hours (minimum); counts Tue–Thu, school in session, less than 12 months old
PHF (future conditions)	0.90 per Standards
Heavy vehicles	Average of duals and TTSTs, 2% minimum
RTOR (future analysis)	Not permitted per Standards
Turn-lane storage	Greater of Synchro 95th percentile and SimTraffic maximum queue, rounded up to 25 ft; 100 ft minimum
Roundabout threshold	v/c 0.85
Study-area rule	Intersections where site trips add 10% or more to background volumes on any approach/movement

2.1 Trip Generation Basis

NCDOT's Best Practices call for the ITE Trip Generation Manual with the Department's Rate-vs-Equation spreadsheet, internal capture per the NCHRP 684 spreadsheet method (vehicle occupancy 1.1, 4,000-ft maximum walking distance), and pass-by limited to retail uses with multi-use pass-by capped at 10% of the adjacent street volume. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal; applied internal-capture and pass-by credits must be reconciled against the NCDOT limits at scoping.

3.0 ANALYSIS CONDITIONS AND MEASURES OF EFFECTIVENESS

Scenarios follow the NCDOT Standards set: Existing Base Year; No-Build Design Year (build-out); and Design Year Build. Where a scoped project is phased, intermediate build-out years are analyzed with earlier-phase trips carried as site trips (never background). Where an impacted STIP project is in planning, design, or construction (or within 5 years post-construction) and site traffic was not in the STIP forecast, an additional STIP design-year analysis is coordinated per the Best Practices TIA/STIP flow chart.

3.1 Intersection MOE Summary (NCDOT format)

Approaches are ordered EB, WB, NB, SB and movements left-through-right per the Best Practices reporting format. Signalized intersections report control delay and LOS overall and per lane group (any v/c > 1.0 is reported as LOS F regardless of delay); unsignalized intersections report no overall LOS — results are per lane group with the conflicting movement. SimTraffic maximum-queue columns are populated at submittal.

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
1	Fox Rd. & Spring Forest Rd.	C / 23.3s	C / 23.6s	C / 26.9s
2	Hollenden Dr. & Spring Forest Rd.	D / 37.9s	D / 39.6s	D / 48.8s

4.0 MITIGATION CRITERIA CHECK (POLICY CH. 5.J)

NCDOT requires improvements when, comparing base-network and project conditions: total average delay increases by 25% or more while remaining at the same LOS; the LOS degrades by one level; or the intersection operates at LOS F. Turn-lane mitigation applies where the 95th-percentile queue exceeds existing storage. Signal-timing changes alone are not considered mitigation. The District Engineer makes the final mitigation determination.

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
1	Fox Rd. & Spring Forest Rd.	+14%	no	no	none indicated
2	Hollenden Dr. & Spring Forest Rd.	+23%	no	no	none indicated

5.0 CONCLUSIONS AND SUBMITTAL NOTES

No study intersection is projected to trigger the Policy Ch. 5.J mitigation criteria under Build conditions at the screening level.

Data conventions: NCDOT forecasts supply AADT, % duals and TTSTs, directional split (D), and design-hour factor (K); forecast conversion to peak-hour volumes uses the Intersection Analysis Utility (IAU) in the NCDOT Traffic Engineering Suite. AADT reference: NCDOT ArcGIS AADT mapping

application. Crash analysis: a site-radius crash assessment from NCDOT's TEAAS (Traffic Engineering Accident Analysis System) is prepared at submittal — not generated by this screening tool.

This TIA is to be prepared under the direct charge of and sealed by a licensed North Carolina Professional Engineer with expertise in traffic engineering (Policy Ch. 5.A; 21 NCAC 56 .1103 — seal with signature over or adjacent to the seal and date of signing; firm name, address, and NC firm license number on the title block). TIAs marked confidential remain so until a driveway permit is requested or the project is publicly announced.

SCREENING DISCLOSURE: this document is a screening-level draft. It does not include the Approved Scoping Document, SimTraffic outputs, Synchro models, TEAAS crash analysis, or signed/sealed certification — all required for a formal NCDOT TIA submittal.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Fox Rd. & Spring Forest Rd.	0.20	0.5	166	63.1%
Hollenden Dr. & Spring Forest Rd.	0.30	0.7	97	36.9%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 94% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 6% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 0.97. Per-signal v/c , delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

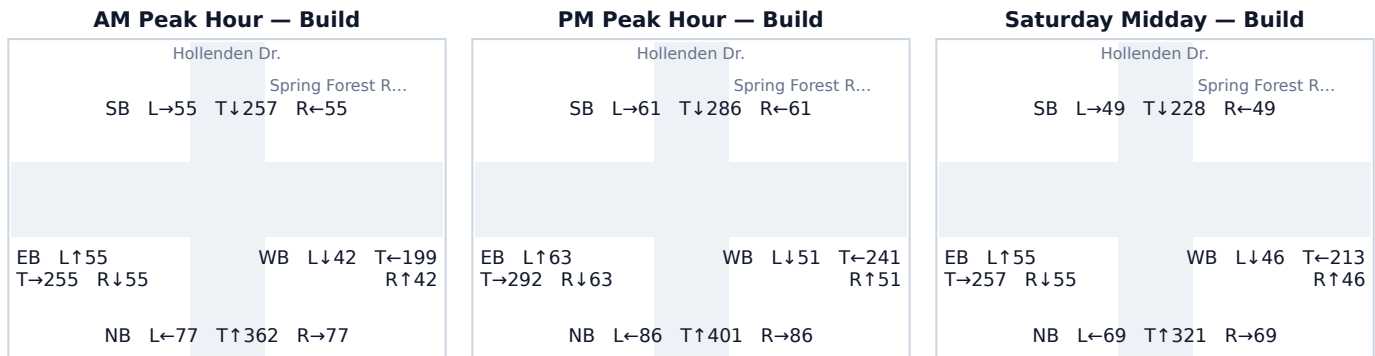
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Hollenden Dr. & Spring Forest Rd.

Signal ID	raleigh-durham-1299179770
Location	SE Raleigh-Durham · 0.30 mi from site
Added PM peak trips	97
Existing / No-Build (PM)	LOS D · 39.6 s/veh · v/c 0.91
Build (PM)	LOS D · 48.8 s/veh · v/c 0.97
Δ control delay	+9.2 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	573	0	573	0.71 → 0.71	25.1 → 25.1	C → C	477
SB	408	0	408	0.50 → 0.50	19.8 → 19.8	B → B	299
EB	371	47	418	0.46 → 0.52	19.0 → 20.1	B → C	308
WB	293	50	343	0.36 → 0.42	17.5 → 18.4	B → B	241

Affected movements (PM peak project trips)

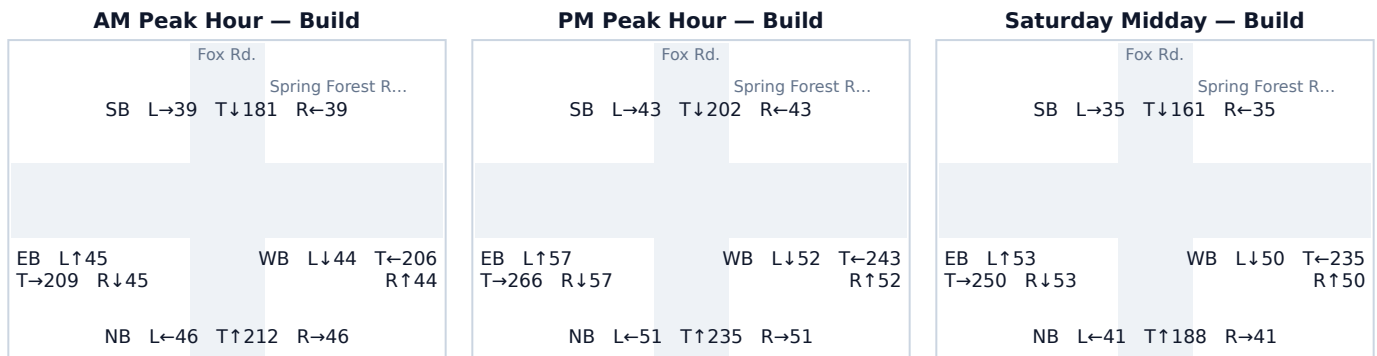
Movement	Project trips	% of project trips
WB Through	50	52%
EB Through	47	48%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Fox Rd. & Spring Forest Rd.

Signal ID	raleigh-durham-195601497
Location	SE Raleigh-Durham · 0.20 mi from site
Added PM peak trips	166
Existing / No-Build (PM)	LOS C · 23.6 s/veh · v/c 0.66
Build (PM)	LOS C · 26.9 s/veh · v/c 0.75
Δ control delay	+3.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	337	0	337	0.42 → 0.42	18.3 → 18.3	B → B	235
SB	288	0	288	0.36 → 0.36	17.4 → 17.4	B → B	195
EB	294	86	380	0.36 → 0.47	17.5 → 19.2	B → B	274
WB	268	80	347	0.33 → 0.43	17.1 → 18.5	B → B	244

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	86	52%
WB Through	80	48%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.